

POL40950: Introduction to Statistics

School of Politics and International Relations, University College Dublin

Autumn Trimester 2026/27 • 10 ECTS

Draft syllabus, version 1. Subject to change before term.

Module Coordinator	Dr Elisa D’Amico
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Office	Room F313, Newman Building
Office hours	Thursdays, 10:00–12:00, online (book a slot)
Class time	Mondays, 13:00–14:50, from 14 September to 23 November. <i>Three Mondays do not meet in the classroom: 7 September is online, 5 October is given over to one-to-one project meetings, and 26 October is the Bank Holiday.</i>
Location	L3.15-LEA , Room L3.15 on Level 3 of the O’Connor Centre for Learning .
What to bring	Your own laptop , with R and RStudio already installed. The room is not a computer lab, so no machines are provided.
Format	Pre-recorded video lessons plus a weekly in-person workshop
Module website	Brightspace

Welcome and module description

Welcome to Introduction to Statistics. This module is the foundational course in the quantitative methods sequence for SPIRe Master’s students, and it is open to PhD students. It assumes no prior background in statistics, mathematics, or coding. If you have never written a line of code or run a regression, you are in the right place, and the module is built to take you from there.

Do droughts make civil conflict more likely? Do wealthier countries fight less often? Does climate-related aid reduce political instability? Are democracies less likely to go to war? Questions like these are answered by collecting and analysing data, and this module gives you the tools to do exactly that. The goal is simple to state: by the end of term you will be able to take social science data, explore and describe it, fit and interpret a regression model, and write up what you found. I teach statistics as something you do rather than something you watch, so the course is hands-on from the start, using the statistical programming language **R**. My in-class examples come from my own field, the study of climate and conflict, but for your own work you will choose a dataset that interests you from a databank I provide, so you spend your time analysing real social science data rather than hunting for it.

There are many statistical tools available to social scientists, but the linear regression model is the most widely used, and a solid understanding of it is required to read or write quantitative research. The module focuses on regression: how to fit it, how to interpret it, and how to reason about statistical inference (how do I know whether my findings hold beyond my sample?). In the final week I situate what you have learned within the wider landscape of quantitative methods (logistic regression, multilevel models, causal inference designs, structural equation modelling, machine learning, and others), so you leave knowing what other tools exist and when you would reach for them.

Learning outcomes

On completion of this module you should have:

1. a basic understanding of working with R and RStudio;
2. the ability to wrangle, summarise, describe, and visualise statistical data;
3. a basic understanding of statistical inference;
4. a basic understanding of executing and interpreting multiple regression;
5. a preliminary understanding of logistic regression.

This module's primary contribution is to UN Sustainable Development Goal 4 (Quality Education). Because the examples and your research papers engage climate and conflict themes, it also touches SDG 13 (Climate Action) and SDG 16 (Peace, Justice and Strong Institutions).

How the module runs

How a typical week works

Most new material is delivered as **short pre-recorded video lessons**. Before each Monday you do two things: watch the week's videos, and take a short **warm-up quiz** that ends by asking what you found most confusing. Our Monday session then runs in three parts:

- *about 15 minutes*: I open by answering the questions you flagged and recapping the trickiest points;
- *about 15–20 minutes*: I cover the harder idea live, with a quick demonstration (more time on weeks where the concept is harder, such as the Central Limit Theorem and inference);
- *the rest of the session*: a hands-on **lab**, where you work through guided R exercises in your pods while I move around the room and help.

Watching the videos beforehand is what makes the lab work, so please keep current.

Labs, assignments, and how they fit together

The **labs** are guided practice and are not graded individually. Each is a Quarto worksheet with step-by-step tasks worked on a shared dataset so you can follow along. A few days after each lab, once you have had a chance to attempt it, I post my clean, commented solution code, so you have a correct reference to check your work against and to adapt for your own analysis. If you do not finish in class, complete it on your own. The labs build the exact skills you then apply, on your *own* chosen dataset, in the graded **milestones** (M1–M3), which in turn become your research paper. So there is no separate pile of homework: the labs are your practice, and the milestones are your project, built in stages.

The weekly rhythm and deadlines

Each teaching week follows the same pattern:

- **Before class (by Saturday 23:59)**: watch the week's videos and submit the short warm-up quiz, including the “most confusing” question. It is graded on completion, not on getting the answers right, so answer honestly. Videos are posted well in advance, so you have plenty of time.
- **Monday 13:00–14:50**: we open with your questions and any loose ends from the previous week, cover the harder idea, then work the lab.
- **During the week**: finish the lab on your own if you did not complete it in class, and check it against the posted solution code. Labs are not submitted, but they are your practice for the milestones, so do not skip them.

The graded deadlines are the three milestones (Fridays of Weeks 5, 9, and 11, at 23:59), the presentation (start of Week 12), and the paper (Friday 11 December). Outside those, the only thing due each week is the warm-up quiz.

Your feedback during the module

Around Week 6 I run a short anonymous survey asking what is working, what is hard, and what I should change, and I tell you in class what I am adjusting as a result. The weekly “most confusing” question does the same job week to week. This is your course as much as mine, and I would much rather hear about a problem in Week 6 than in the end-of-term review.

Two online weeks, and one week with no class

Three Mondays do not meet in the room, so put them in your calendar now:

- **Monday 7 September (Week 1) — online, no live class.** Your onboarding week.
- **Monday 5 October (Week 5) — no lecture.** Instead everyone books a **10-minute project meeting** with me: **Monday 5 October, 13:00–18:00** or **Tuesday 6 October, 09:30–14:30**. These are **on Zoom**, not in the classroom; the link comes with your booking confirmation.
- **Monday 26 October (Week 8) — no class at all**, for the October Bank Holiday.

Week 1 is asynchronous: nothing happens at 13:00, and you work through pre-recorded content and a short task in your own time. **Week 5** has no lecture either, but it is not a week off from me: everyone books a **10-minute one-to-one project meeting** on Zoom (Monday 5 or Tuesday 6 October) to talk through their research question before Milestone 1 is due on the Friday. Everything you need is posted in advance, and Milestone 1 is still due in Week 5 as normal. Week 8 has no class and no new material; use it to catch up and work on your project.

Support pods

Early in the term you will submit a few topics that interest you, and I will place you into small **support pods** (about four students each) grouped by shared interest. Pods are for helping one another through the labs and the staged assignments. Your written work, your presentation, and your grade remain entirely your own.

Communication

Course questions, and especially questions about R code and error messages, go to the “**Code and Errors**” **discussion forum on Brightspace**. Please search the forum and the error text first, then post. You are encouraged to answer one another, and I will be in the conversation as well. I do not use Slack for this module. For private matters, email me.

Getting unstuck with R

Hitting errors is a normal part of coding, not a sign you are doing badly. When R will not cooperate, work through these steps before asking, since they solve most problems and build the exact skill you are here to learn:

1. Search the relevant chapter of ModernDive or K&W for the function or idea you are stuck on.
2. Put the problem in your own words and search it online, adding **rstats** or **tidyverse** to the query (for example, “how to import a csv in R tidyverse”).
3. Copy the exact text of the error message and search that. Someone has almost certainly seen it before.
4. If you are still stuck, post to the Code and Errors forum with your code and the error. A classmate or I will help, and your question will help the next person too.

Software

We use **R** and **RStudio**, both free, and you will write your work in **Quarto** documents (these let you combine code, output, and writing in one file and render it to a clean report). I recommend RStudio rather than other editors, since everything you need is in one place and all course materials assume it.

- Download R: <https://cran.r-project.org/>
- Download RStudio (free desktop version): <https://posit.co/download/rstudio-desktop/>
- Quarto (bundled with recent RStudio): <https://quarto.org/>

Bring your own laptop to every class. Our teaching room is not a computer lab, so no machines are provided, and the Monday session is hands-on from the start. Install R and RStudio before Week 2; the Week 1 onboarding walks you through it step by step and includes a short task to prove your install works, so any problems surface early rather than in class. If you cannot install on your own machine, or your laptop gives up mid-term, you can run everything free in your browser through **Posit Cloud** (<https://posit.cloud/>), which needs nothing installed. Tell me if you are in this position and I will make sure you are set up. Templates for every assignment are provided, so you never start from a blank page.

Texts and resources

The module uses two core texts, both free to you: **K&W** for the concepts and **ModernDive** for the R. A third, **QSS**, is an optional further reference.

- Kellstedt, Paul, and Guy Whitten. 2018. *The Fundamentals of Political Science Research*, 3rd edition. Cambridge University Press. (**K&W**, core; free online through the UCD Library)
- Ismay, Chester, Albert Y. Kim, and Arturo Valdivia. 2025. *Statistical Inference via Data Science: A ModernDive into R and the Tidyverse*, 2nd edition. CRC Press. Free online: <https://moderndive.com/v2/> (**ModernDive**, core for the R)
- Imai, Kosuke. 2017. *Quantitative Social Science: An Introduction*. Princeton University Press. (**QSS**, optional further reading; UCD Library print copy)

You do not need to buy these. K&W is free online through the UCD Library, ModernDive is free online at the link above, and QSS is available as a print copy in the UCD Library. For most weeks, K&W and ModernDive, both freely accessible to all of you, cover the core material, with QSS offered as a further explanation, so you are never dependent on the library print copy. I will assign specific chapters week by week and post all other materials on Brightspace.

Further resources (optional)

None of these are required. They are here for anyone who wants another explanation, a deeper treatment, or a push beyond the module, and several are excellent for the research paper. All of the books below are free online.

- *Statistical thinking*: D. Spiegelhalter, *The Art of Statistics* (Penguin, an accessible read on what statistics is really for).
- *Research design and causal inference*: N. Huntington-Klein, *The Effect*, <https://theeffectbook.net>; S. Cunningham, *Causal Inference: The Mixtape*, <https://mixtape.scunning.com>.
- *More R*: Wickham, Çetinkaya-Rundel, and Golemund, *R for Data Science*, <https://r4ds.hadley.nz>; D. Navarro, *Learning Statistics with R*, <https://learningstatisticswithr.com>; Larsen and Fazekas, *Quantitative Politics with R*, <https://qpplr.com>.
- *Data visualisation*: K. Healy, *Data Visualization*, <https://socviz.co>; C. Wilke, *Fundamentals of Data Visualization*, <https://clauswilke.com/dataviz>.
- *Video*: StatQuest with Josh Starmer (YouTube) for statistical intuition; Nick Huntington-Klein (YouTube) for R and causal inference.

Assessment

There is no exam in this module. You build a short individual research paper across the term, in stages, with feedback at each step, and you present it. The components are below.

Component	Weight	Timing
Participation in learning activities	10%	Throughout (Weeks 1–12)
Staged assignments (three milestones)	35%	Fri 9 Oct (Wk 5), Fri 6 Nov (Wk 9), Fri 20 Nov (Wk 11)
Topic proposal (credited under participation)	–	Fri 25 Sep (Wk 3)
Research paper (individual, ~3,000 words)	45%	Fri 11 Dec (after teaching)
Presentation (recorded video)	10%	Sun 22 Nov (start of Wk 12)
Total	100%	

Participation in learning activities (10%)

This rewards staying current with the module. It is made up of the Week 1 readiness check, the **topic proposal** (Week 3), the short weekly **warm-up quizzes** (from Week 2 onwards: a few quick questions on that week’s videos, ending by asking what you found most confusing), the Week 5 project meeting, the mid-semester check-in, and the **Week 12 peer feedback** you write on a classmate’s presentation. Everything here is graded on **completion, not on getting the answers right**, so answer honestly. There is no penalty for a wrong answer, and seeing what the class finds hard is how I decide what to go over in class. (The mid-semester check-in is anonymous; you get the credit simply for completing it.) The point is to keep you watching the videos and working steadily, which is what this format depends on.

Staged assignments / milestones (35%)

Three R-based assignments, each building one section of your research paper. You submit each as a rendered Quarto document on Brightspace, and I return it with feedback before the next is due, so you improve as you go.

- **Topic proposal** (due Fri 25 Sep, Week 3): a short note on the question and dataset you are thinking of. This is the first step of the sequence but it is **not part of the 35%**; it is credited under participation, on completion. Its job is to get you thinking early and to let me flag anything unworkable before you invest in it.
- **Milestone 1** (due Fri 9 Oct, Week 5): research question, your chosen dataset (from the databank or your own), and descriptive analysis. This is also your topic **clearance**: I sign off on your question and data before you proceed.
- **Milestone 2** (due Fri 6 Nov, Week 9): regression analysis and interpretation.
- **Milestone 3** (due Fri 20 Nov, Week 11): full draft, with any extension (categorical predictors, interactions, or logistic regression) that suits your question.

Research paper (45%)

An individual paper of about 3,000 words (excluding references, tables, and figure captions) that answers a research question using regression analysis. You choose a dataset from a **databank I provide on Brightspace** (available from Week 1, so you can browse early): a set of clean, ready-to-use datasets covering a range of social science topics, so you can pick something that interests you without having to find and clean data yourself. (My in-class examples use climate and conflict data, but you are not tied to that topic.) If you would rather use your own data, you are welcome to: flag it when you submit your topics in Week 3 so I can check it is feasible, and it is confirmed at Milestone 1. **If you plan ahead, this paper can become a piece of your thesis or dissertation.** You are encouraged, PhD students especially, to use a research question and data from your own research. If the data you want is not readily available, contact me to run alternatives by me, or schedule a meeting. The datasets in the Data Bank are all cross-sectional (one row per case), which is what this module teaches. If you already have a

stronger background in statistics, you are welcome to take on more advanced data, for example panel or time-series data, using appropriate methods, as long as you clear it with me first. The paper should cover: the research question and why it matters; what previous work and theory lead you to expect; your data and method; your results; and a short conclusion. Use APSA referencing throughout. Submit as a PDF on Brightspace by **Friday 11 December 2026, 23:59**.

Your finished paper and its R code together make a **reproducible, professional analysis** you can add to a portfolio or show employers. Particularly strong papers may also be invited to present at a School research event or postgraduate showcase.

The deadline is deliberately after teaching ends and inside the examination period, so you have the fortnight after our last class to write without other coursework competing for it. If you have exams that fortnight, plan backwards from 11 December accordingly.

Bonus: helping on the forum

Good answers on the **Code & Errors forum** earn bonus marks on your final paper. For each coding question where you post a correct, genuinely helpful answer that resolves a classmate's problem, you earn **half a mark** added to your paper grade, up to a maximum of **5 marks**. I confirm which answers count (a clear, correct fix, not a "me too" or a near miss). This rewards the people who make this a supportive place to learn, and in practice help often reaches you faster than I can get to it. This is bonus only: it can raise your paper mark but never lowers it.

Presentation (10%)

A five-minute recorded video in which you present and interpret your own analysis on camera, due at the start of Week 12. Presenting is a normal part of doing research, and it lets you show your understanding in your own words. We use these videos in the **Week 12 peer feedback workshop**. Each of you is randomly assigned a peer's presentation to review using a short feedback form, and we then discuss the feedback in your pods during the Week 12 session. You use what you receive to strengthen your paper before the final deadline.

Collaboration

You are encouraged to help one another, talk through problems, and learn together in your pods. The work you submit must be your own: your code, your analysis, your interpretation, and your writing.

Feedback

Feedback is provided individually within 20 working days of submission, per university guidelines. Because the milestones feed directly into the paper, the feedback is designed to be used.

Grades

The module is graded on the UCD letter-grade scale. The indicative percentage-to-grade conversion is below; confirm current bands with the School Office.

Mark	Grade	Mark	Grade
97–100	A+	71–73	D+
94–96	A	68–70	D
91–93	A–	65–67	E+
88–90	B+	54–64	E
85–87	B	44–53	E–
83–84	B–	0–43	F
80–82	C+		
77–79	C		
74–76	C–		

Remediation

If you do not pass, the resit is a resubmission of the research paper, within two trimesters (Spring). There is no terminal exam for this module.

Use of generative AI

Generative AI tools (such as ChatGPT, Claude, and GitHub Copilot) are part of how people now write and debug code, and I am not going to pretend otherwise. You may use them in this module to help you **learn concepts, understand error messages, and debug your R code**. You may not use them to write your analysis, your interpretation, or your paper for you. The thinking and the writing must be yours.

To make the line concrete:

- **Fine** (just disclose it): “what does this error message mean?”; “why will my code not run?”; “explain a confidence interval in plain language”; “how do I add a regression line to a ggplot?”; “what is the difference between `filter()` and `select()`?”
- **Not allowed:** asking it to write your analysis, interpretation, results, or paper; pasting an assignment prompt and submitting what it returns; handing in AI-written text or code you cannot explain in your own words.

The test is simple: AI may help you *learn how*; the analysis, the judgement, and the writing must be yours.

One module-specific rule on timing: **keep AI tools closed during class and labs**. That is your space to build the skill by doing it yourself, with me and your classmates on hand to help, which is better than any AI. Outside class, on your own project work, the uses above are fine, logged. I will not police laptops; this works on trust, and the milestones and recorded presentation are what show whether the ability is genuinely yours.

All use of AI must be logged and disclosed, on every submission. Every quiz and assignment in this module has a built-in place to record your AI use, so disclosing is easy and routine:

- **Milestones and the paper:** the Quarto template includes an **AI-use log** section at the end, and the Brightspace dropdown also accepts a separate log file.
- **Quizzes:** each ends with a short AI-use declaration.
- **If you did not use AI, simply say so** (“No AI used”). A one-line entry is fine.

When you do use a tool, your log should record:

1. the tool and version used (for example, “ChatGPT, GPT-x, accessed 3 Nov 2026”);
2. the prompts you gave it; and
3. how you used the output, and what you changed.

A short, honest log is all I am asking for, and disclosed use that follows these rules is not penalised. Be aware that it is usually easy to tell when AI has been used and no log has been attached: submitted work that does not match your demonstrated ability, generic or formulaic phrasing, code you cannot explain, and fabricated or incorrect references and results are all giveaways. The recorded presentation is also there to confirm that the analysis is genuinely yours. **Undisclosed AI use (using AI but submitting no log) will be treated as academic misconduct** under UCD policy. If you are unsure whether something is allowed, disclose it, or ask me.

Module policies

Extensions and deadlines

From September 2026, UCD's [Additional Considerations for Assessment](#) policy replaces the previous late submission and extenuating circumstances policies. Three things follow for this module.

Self-declare extension. You may apply through SISWeb, in advance of the deadline, for five extra working days. No reason and no approval are needed. In this module it is available for the three milestones and the research paper. It is not available for the recorded presentation, or for the participation components, which run on rolling weekly deadlines. The further five days with a one-point grade penalty is not enabled here, so five working days is the maximum available by self-declaration.

Longer than that. Anything beyond ten working days is handled through Certified Additional Considerations, which require supporting documentation and are assessed by the College Office rather than by me.

Missing a deadline without applying results in a grade of ABS for that component. Applying in advance costs you nothing, so if you think you may need the time, apply before the deadline rather than after it.

If something is going wrong, tell me early. I cannot grant extensions myself under the new policy, but the sooner I know, the more I can help you plan around it.

Academic integrity and plagiarism

Plagiarism, presenting someone else's work, words, or ideas as your own, is taken seriously at UCD and in this module. Please read the UCD guidance: <https://libguides.ucd.ie/academicintegrity>. Academic misconduct is handled under School, College, and University procedures. The rules on generative AI above are part of this module's integrity expectations.

Inclusion and accessibility

I want this module to work for everyone. If you have a disability or a condition that affects your participation or assessment, please contact UCD Access and Lifelong Learning (<https://www.ucd.ie/all/>) and let me know early so we can arrange reasonable accommodations.

Withdrawal

UCD's automatic withdrawal period (with an automatic grade of W) runs to six weeks from the start of term. After that, withdrawals are granted only for good reason; a poor assignment score is not sufficient reason.

Recordings and copyright

Materials I provide, including videos, slides, handouts, datasets, and assignment briefs, are for your use in this module and are protected by copyright. Please do not redistribute them without my permission.

Keep copies of your work

Please keep a copy of everything you submit until your final grade has been released, and retain returned, graded work until then as well.

References and recommendations

I am happy to write reference letters (for further study, internships, or jobs) for students who have done well in the module, normally those achieving at least an A–. I write references only after the module is complete. Please give me plenty of notice, and note that letters are usually sent directly to the organisation rather than to you.

Student effort

This is a 10 ECTS module. Indicative effort: 12 hours of lectures (delivered as video), 12 hours of computer-aided lab/workshop, and around 200 hours of autonomous learning (watching the videos, reading, the labs, and the research paper), for roughly 224 hours in total.

Changes to this syllabus

I reserve the right to make reasonable changes to this syllabus during the term, for example adjusting the pace or the readings to fit the class. Any changes will be announced in class and posted to Brightspace.

Course schedule

Teaching runs from Monday 7 September to Monday 23 November 2026. Each week has pre-recorded videos to watch beforehand and a Monday workshop, except Week 1, which is asynchronous, Week 5, which replaces the lecture with one-to-one project meetings, and the week of 26 October, when there is no class for the October Bank Holiday. Each week's core readings are from K&W and ModernDive (both free to you); QSS gives an optional parallel treatment of most topics if you want another explanation. Exact sections are confirmed on Brightspace. The schedule is indicative and I may adjust it as we go; any change is announced in class and on Brightspace. All assignment deadlines are collected in the Assessment section above.

Week 1 (7 September): Getting set up (asynchronous, no live class)

Online. No live class, and nothing at 13:00.

This is your onboarding week. Install R and RStudio, watch the welcome and installation videos, and complete the Readiness Check: a short background survey, proof that your install works, and a quick syllabus quiz. Take a first look at the dataset databank too, and jot down a topic or two that interest you, so you are ready to choose in Week 3. Post any questions to the discussion forum so you arrive in Week 2 ready to work.

Read: ModernDive ch. 1; K&W ch. 1. *Due:* Week 1 Readiness Check.

Week 2 (14 September): What is quantitative social science? Data and visualisation

Room L3.15, O'Connor Centre for Learning (Level 3). Monday 13:00–14:50.

We start with the basic vocabulary: cases, variables, levels of measurement, and what a dataset actually is. You tour RStudio, import the climate-conflict dataset, and make your first plots. By the end you can load data and describe it visually.

Read: K&W ch. 6; ModernDive ch. 2 and ch. 4. *In class:* data detective; import and ggplot walkthrough.

Week 3 (21 September): Describing data

Room L3.15, O'Connor Centre for Learning (Level 3). Monday 13:00–14:50.

How to summarise variables numerically: centre (mean, median), spread (variance, standard

deviation), and shape, with each summary tied to a plot. You also submit a few topics that interest you, which I use to form your support pods.

Read: K&W ch. 5; ModernDive ch. 3. *In class:* summaries and plots; “guess the distribution”.
Due: interest topics. Milestone 1 set.

Week 4 (28 September): Simple regression

Room L3.15, O’Connor Centre for Learning (Level 3). Monday 13:00–14:50.

The core of the module begins. We treat regression first as description: a line that summarises how one variable moves with another. You fit your first model with `lm()`, read the slope and intercept, and meet the Quarto template you will use for assignments.

Read: K&W ch. 9; ModernDive ch. 5. *In class:* build and interpret a regression; interpretation challenge.

Week 5 (5 October): Multiple regression (no lecture — project meetings)

Online. No live class, and nothing at 13:00.

No lecture this week. Watch the two multiple-regression videos and work through the lab on your own, using the forum for questions. **Your 10-minute project meeting** is this week, Monday 5 or Tuesday 6 October, on Zoom. It counts towards participation, and it is your chance to talk through your question before Milestone 1 is due on Friday. Multiple regression lets us hold other variables constant and ask what each one adds. Milestone 1 is also due, and it doubles as your topic clearance: I sign off on your question and data before you go further.

Read: K&W ch. 10; ModernDive ch. 6. *Async:* guided self-check lab. *Due:* **Milestone 1** (topic clearance).

Week 6 (12 October): Model building, prediction, and causation

Room L3.15, O’Connor Centre for Learning (Level 3). Monday 13:00–14:50.

Back in person. We work on building sensible models: which variables to include, the difference between prediction and a causal claim, and how to read a model with proper scepticism. I return Milestone 1 feedback and we run a clinic on your own questions and data.

Read: K&W ch. 3 and ch. 11. *In class:* regression clinic on your own question; short anonymous mid-term feedback survey.

Week 7 (19 October): Sampling and the Central Limit Theorem

Room L3.15, O’Connor Centre for Learning (Level 3). Monday 13:00–14:50.

We shift from describing our data to asking what it tells us about the world. What is a sampling distribution, why does the normal distribution keep appearing, and what is the Central Limit Theorem? This is one of the harder ideas, so expect more live explanation.

Read: K&W ch. 7; ModernDive ch. 7–8. *In class:* sampling simulation; concept polls. *Due:* Milestone 2 set.

Week of 26 October: No class (October Bank Holiday)

No Monday class this week (public holiday). Use the time to catch up, consolidate, and work on your project.

Week 9 (2 November): Statistical inference: hypothesis tests and confidence intervals

Room L3.15, O’Connor Centre for Learning (Level 3). Monday 13:00–14:50.

How do we know whether a relationship in our sample holds more broadly? We cover p-values and confidence intervals and, just as much, how to interpret them honestly and what they do not mean.

Read: K&W ch. 8; ModernDive ch. 9–10. *In class:* interpreting p-values and confidence intervals; misconception polls. *Due:* **Milestone 2**.

Week 10 (9 November): Categorical predictors

Room L3.15, O'Connor Centre for Learning (Level 3). Monday 13:00–14:50.

How to include categorical independent variables (dummy variables) in a regression, and how to interpret the contrasts between categories.

Read: K&W ch. 11; ModernDive ch. 6 (categorical predictors). *In class:* dummy variables in practice. *Due:* Milestone 3 set.

Week 11 (16 November): Interactions and writing up results

Room L3.15, O'Connor Centre for Learning (Level 3). Monday 13:00–14:50.

When the effect of one variable depends on another (interactions), and how to present regression results clearly in tables and prose. We workshop the results paragraph you will use in your paper.

Read: K&W ch. 11; ModernDive ch. 11. *In class:* interpreting interactions; results-paragraph workshop. *Due:* **Milestone 3** (full draft).

Week 12 (23 November): Logistic regression, the methods landscape, and peer feedback

Room L3.15, O'Connor Centre for Learning (Level 3). Monday 13:00–14:50.

Logistic regression (delivered as video) for binary outcomes, such as whether a conflict begins. In class I give a quick tour of where to go next, multilevel models, causal inference designs, structural equation modelling, machine learning, and more, and when each is used. We then run the peer feedback workshop on your presentations.

Read: K&W ch. 12. *Async video:* logistic regression. *In class:* methods tour; peer feedback workshop. *Due:* **Presentation** (recorded).

After teaching

Research paper due Friday 11 December 2026, 23:59.
